

EE263 Homework 5

Summer 2026

1. Identifying a system from input/output data. We consider the standard setup:

$$y = Ax + v,$$

where $A \in \mathbb{R}^{m \times n}$, $x \in \mathbb{R}^n$ is the input vector, $y \in \mathbb{R}^m$ is the output vector, and $v \in \mathbb{R}^m$ is the noise or disturbance. We consider here the problem of estimating the matrix A , given some input/output data. Specifically, we are given the following:

$$x^{(1)}, \dots, x^{(N)} \in \mathbb{R}^n, \quad y^{(1)}, \dots, y^{(N)} \in \mathbb{R}^m.$$

These represent N samples or observations of the input and output, respectively, possibly corrupted by noise. In other words, we have

$$y^{(k)} = Ax^{(k)} + v^{(k)}, \quad k = 1, \dots, N,$$

where $v^{(k)}$ are assumed to be small. The problem is to estimate the (coefficients of the) matrix A , based on the given input/output data. You will use a least-squares criterion to form an estimate $\hat{A}$ of A . Specifically, you will choose as your estimate $\hat{A}$ the matrix that minimizes the quantity

$$J = \sum_{k=1}^N \|Ax^{(k)} - y^{(k)}\|^2$$

over A .

- a) Explain how to do this. If you need to make an assumption about the input/output data to make your method work, state it clearly. You may want to use the matrices $X \in \mathbb{R}^{n \times N}$ and $Y \in \mathbb{R}^{m \times N}$ given by

$$X = \begin{bmatrix} x^{(1)} & \dots & x^{(N)} \end{bmatrix}, \quad Y = \begin{bmatrix} y^{(1)} & \dots & y^{(N)} \end{bmatrix}$$

in your solution.

- b) On the course web site you will find some input/output data for an instance of this problem in the file `sysid_data.json`. Executing this Julia file will assign values to m , n , and N , and create two matrices that contain the input and output data, respectively. The $n \times N$ matrix variable `X` contains the input data $x^{(1)}, \dots, x^{(N)}$ (i.e., the first column of `X` contains $x^{(1)}$, etc.). Similarly, the $m \times N$ matrix `Y` contains the output data $y^{(1)}, \dots, y^{(N)}$. You must give your final estimate $\hat{A}$, your source code, and also give an explanation of what you did.

2. Simultaneous left inverse of two matrices. Consider a system where

$$y = Gx, \quad \tilde{y} = \tilde{G}x$$

where $G \in \mathbb{R}^{m \times n}$, $\tilde{G} \in \mathbb{R}^{m \times n}$. Here x is some variable we wish to estimate or find, y gives the measurements with some set of (linear) sensors, and $\tilde{y}$ gives the measurements with some

alternate set of (linear) sensors. We want to find a *reconstruction matrix* $H \in \mathbb{R}^{n \times m}$ such that $HG = H\tilde{G} = I$. Such a reconstruction matrix has the nice property that it recovers x perfectly from *either* set of measurements (y or $\tilde{y}$), *i.e.*, $x = Hy = H\tilde{y}$. Consider the specific case

$$G = \begin{bmatrix} 2 & 3 \\ 1 & 0 \\ 0 & 4 \\ 1 & 1 \\ -1 & 2 \end{bmatrix}, \quad \tilde{G} = \begin{bmatrix} -3 & -1 \\ -1 & 0 \\ 2 & -3 \\ -1 & -3 \\ 1 & 2 \end{bmatrix}.$$

Either find an explicit reconstruction matrix H , or explain why there is no such H .

3. Optimal flow on a data collection network. We consider a communications network with m nodes, plus a special destination node, and n communication links. Each communication link connects two (distinct) nodes and is bidirectional, *i.e.*, information can flow in either direction. We will assume that the network is connected, *i.e.*, there is a path, or sequence of links, from every node (including the special destination node) to every other node. With each communication link we associate a directed arc, which defines the direction of information flow that we will call positive. Using these reference directions, the flow or traffic on link j is denoted f_j . (The units are bits per second, but that won't matter to us.) The traffic on the network (*i.e.*, the flow in each communication link) is given by a vector $f \in \mathbb{R}^n$. A small example is shown in part 2 of this problem. In this example, nodes 1 and 3 are connected by communication link 4, and the associated arc points from node 1 to node 3. Thus $f_4 = 12$ means the flow on that link is 12 (bits per second), from node 1 to node 3. Similarly, $f_4 = -3$ means the flow on link 4 is 3 (bits per second), from node 3 to node 1. External information enters each of the m regular nodes and flows across links to the special destination node. In other words, the network is used to collect information from the nodes and route it through the links to the special destination node. (That explains why we call it a data collection network.) At node i , an external information flow s_i (which is nonnegative) enters. The vector $s \in \mathbb{R}^m$ of external flows is sometimes called the *source vector*. Information flow is conserved. This means that at each node (except the special destination node) the sum of all flows entering the node from communication links connected to that node, plus the external flow, equals the sum of the flows leaving that node on communication links. As an example, consider node 3 in the network of part 2. Links 4 and 5 enter this node, and link 6 leaves the node. Therefore, flow conservation at node 3 is given by

$$f_4 + f_5 + s_3 = f_6.$$

The first two terms on the left give the flow entering node 3 on links 4 and 5; the last term on the left gives the external flow entering node 3. The term on the righthand side gives the flow leaving over link 6. Note that this equation correctly expresses flow conservation regardless of the signs of f_4 , f_5 , and f_6 . Finally, here is the problem.

- a) The vector of external flows, $s \in \mathbb{R}^m$, and the network topology, are given, and you must find the flow f that satisfies the conservation equations, and minimizes the mean-square traffic on the network, *i.e.*,

$$\frac{1}{n} \sum_{j=1}^n f_j^2.$$

Your answer should be in terms of the external flow s , and the *node incidence matrix* $A \in \mathbb{R}^{m \times n}$ that describes the network topology. The node incidence matrix is defined as

$$A_{ij} = \begin{cases} 1 & \text{arc } j \text{ enters (or points into) node } i \\ -1 & \text{arc } j \text{ leaves (or points out of) node } i \\ 0 & \text{otherwise.} \end{cases}$$

Note that each row of A is associated with a node on the network (not including the destination node), and each column is associated with an arc or link.

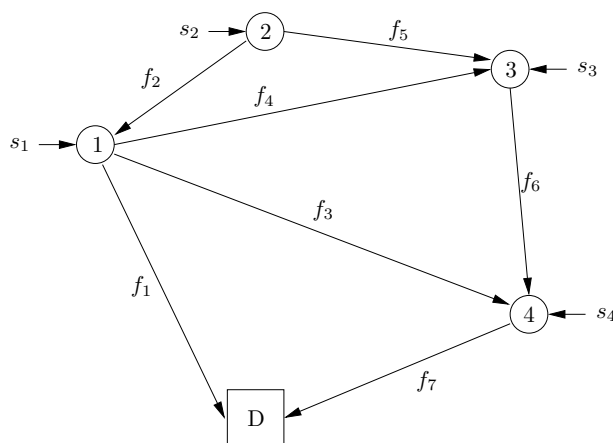
- b) Now consider the specific (and very small) network shown below. The nodes are shown as circles, and the special destination node is shown as a square. The external flows are

$$s = \begin{bmatrix} 1 \\ 4 \\ 10 \\ 10 \end{bmatrix}.$$

One simple feasible flow is obtained by routing all the external flow entering each node along a shortest path to the destination. For example, all the external flow entering node 2 goes to node 1, then to the destination node. For node 3, which has two shortest paths to the destination, we arbitrarily choose the path through node 4. This simple routing scheme results in the feasible flow

$$f_{\text{simple}} = \begin{bmatrix} 5 \\ 4 \\ 0 \\ 0 \\ 0 \\ 10 \\ 20 \end{bmatrix}.$$

Find the mean square optimal flow for this problem (as in part 1). Compare the mean square flow of the optimal flow with the mean square flow of f_{simple} .



4. Optimal control of unit mass. In this problem you will use the language you prefer to solve an optimal control problem for a force acting on a unit mass. Consider a unit mass at position $p(t)$ with velocity $\dot{p}(t)$, subjected to force $f(t)$, where $f(t) = x_i$ for $i - 1 < t \leq i$, for $i = 1, \dots, 10$.

- a) Assume the mass has zero initial position and velocity: $p(0) = \dot{p}(0) = 0$. Find x that minimizes

$$\int_{t=0}^{10} f(t)^2 dt$$

subject to the following specifications: $p(10) = 1$, $\dot{p}(10) = 0$, and $p(5) = 0$. Plot the optimal force f , and the resulting p and $\dot{p}$. Make sure the specifications are satisfied. Give a short intuitive explanation for what you see.

- b) Assume the mass has initial position $p(0) = 0$ and velocity $\dot{p}(0) = 1$. Our goal is to bring the mass near or to the origin at $t = 10$, at or near rest, *i.e.*, we want

$$J_1 = p(10)^2 + \dot{p}(10)^2,$$

small, while keeping

$$J_2 = \int_{t=0}^{10} f(t)^2 dt$$

small, or at least not too large. Plot the optimal trade-off curve between J_2 and J_1 . Check that the end points make sense to you. *Hint:* the parameter μ has to cover a very large range, so it usually works better in practice to give it a logarithmic spacing. You don't need more than 50 or so points on the trade-off curve.

Your solution to this problem should consist of a clear written narrative that explains what you are doing, and gives formulas symbolically, and the final plots produced.

5. Modifying measurements to satisfy known conservation laws. A vector $y \in \mathbb{R}^n$ contains n measurements of some physical quantities $x \in \mathbb{R}^n$. The measurements are good, but not perfect, so we have $y \approx x$. From physical principles it is known that the quantities x must satisfy some linear equations, *i.e.*,

$$a_i^\top x = b_i, \quad i = 1, \dots, m,$$

where $m < n$. As a simple example, if x_1 is the current in a circuit flowing into a node, and x_2 and x_3 are the currents flowing out of the node, then we must have $x_1 = x_2 + x_3$. More generally, the linear equations might come from various conservation laws, or balance equations (mass, heat, energy, charge ...). The vectors a_i and the constants b_i are known, and we assume that $a_1, \dots, a_m$ are independent. Due to measurement errors, the measurement y won't satisfy the conservation laws (*i.e.*, linear equations above) exactly, although we would expect $a_i^\top y \approx b_i$. An engineer proposes to adjust the measurements y by adding a correction term $c \in \mathbb{R}^n$, to get an adjusted estimate of x , given by

$$y_{\text{adj}} = y + c.$$

She proposes to find the smallest possible correction term (measured by $\|c\|$) such that the adjusted measurements y_{adj} satisfy the known conservation laws. Give an explicit formula for

the correction term, in terms of y , a_i , b_i . If any matrix inverses appear in your formula, explain why the matrix to be inverted is nonsingular. Verify that the resulting adjusted measurement satisfies the conservation laws, *i.e.*, $a_i^\top y_{\text{adj}} = b_i$.

6. (Optional) Invertibility of certain matrices. In this problem you will show that, for *any* matrix A , and any positive number μ , the matrices $A^\top A + \mu I$ and $AA^\top + \mu I$ are both invertible, and

$$(A^\top A + \mu I)^{-1} A^\top = A^\top (AA^\top + \mu I)^{-1}.$$

a) Let's first show that $A^\top A + \mu I$ is invertible, assuming $\mu > 0$. (The same argument, with $A^\top$ substituted for A , will show that $AA^\top + \mu I$ is invertible.) Suppose that $(A^\top A + \mu I)z = 0$. Multiply on the left by $z^\top$, and argue that $z = 0$. This is what we needed to show. (Your job is to fill all details of the argument.)

b) Now let's establish the identity above. First, explain why

$$A^\top (AA^\top + \mu I) = (A^\top A + \mu I) A^\top$$

holds. Then, multiply on the left by $(A^\top A + \mu I)^{-1}$, and on the right by $(AA^\top + \mu I)^{-1}$. (These inverses exist, by part (a).)

c) Now assume that A is fat and full rank. Show that as μ tends to zero from above (*i.e.*, μ is positive) we have

$$(A^\top A + \mu I)^{-1} A^\top \rightarrow A^\top (AA^\top)^{-1}.$$

(This is asserted, but not shown, in the lecture notes on page 8-12.)

7. Invertibility of the KKT matrix. Recall the linearly-constrained norm minimization problem we saw in lecture: Given $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{k \times n}$, and $d \in \mathbb{R}^k$, find $x \in \mathbb{R}^n$ to

$$\begin{aligned} &\text{minimize} && \|Ax - b\| \\ &\text{subject to} && Cx = d. \end{aligned}$$

We assume that $k \leq n$, so there are at least as many variables as equality constraints. As discussed in lecture, using the theory of constrained optimization, a necessary condition for a point $x \in \mathbb{R}^n$ to be a minimizer of this problem is for there to exist $\lambda \in \mathbb{R}^k$ (the Lagrange multiplier) so that x, λ solve:

$$\begin{bmatrix} A^\top A & C^\top \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} A^\top b \\ 0 \end{bmatrix}.$$

This system is called the Karush-Kuhn-Tucker (KKT) system, and the matrix

$$K = \begin{bmatrix} A^\top A & C^\top \\ C & 0 \end{bmatrix} \in \mathbb{R}^{(n+k) \times (n+k)}$$

is called the KKT matrix. In this problem you will fill in the proof that K is invertible if and only if C has linearly independent rows, and

$$S = \begin{bmatrix} A \\ C \end{bmatrix}$$

has linearly independent columns. Notice that if $A = I$, then S always has linearly independent columns and K is invertible if and only if C has linearly independent rows (which covers the case of the least norm problem studied in lecture).

- a) Show that $\text{null}(A) \cap \text{null}(C) = \{0\}$ if and only S has linearly independent columns.
- b) Show that if C does not have linearly independent rows, then K is singular (i.e., not invertible).
- c) Show that if there exists a nonzero vector $u \in \text{null}(A) \cap \text{null}(C)$, then K is singular.
- d) Show that if K is singular, then either C does not have linearly independent rows, or there exists a nonzero vector $u \in \text{null}(A) \cap \text{null}(C)$.